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Bwrdd Iechyd Prifysgol
Betsi Cadwaladr
University Health Board

ANAESTHESIA, CRITICAL CARE & PAIN MANAGEMENT DIRECTORATE

OBSTETRIC ANAESTHESIA WRITTEN CONTROL DOCUMENT

CARDIOPULMONARY RESUSCITATION IN PREGNANCY

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Author(s):	Dr Gareth Roberts	**Author(s) title:**	Consultant Anaesthetist
Responsible dept / director:	Department of Anaesthesia, BCUHB (West) Dr Linda Warnock Clinical Lead of Anaesthesia, BCUHB (West)		
Approved by:	Department of Anaesthesia, BCUHB (West)		
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1. Introduction and Purpose

Cardiac arrest in pregnancy is thought to occur in approximately 1:30,000 maternities[1]. Cardiac arrest during late pregnancy or delivery is even rarer. However, when it does occur maternal and fetal survival rates are low. Exact figures are not known but it has been suggested that the figure for maternal survival is around 40% [2]. This is at least partly because the events leading to cardiac arrest tend to be overwhelming and incurable, but another factor of major importance is that the physiological changes of pregnancy hamper resuscitative efforts.

2. Scope

This guideline is intended as a reference for all anaesthetists, obstetricians & midwives.

3. Points to consider

3.1 Best Practice

- All staff involved in intrapartum care should be familiar with current basic life support guidelines for the pregnant patient and should follow them during resuscitation attempts
- 30 degree left lateral tilt should be used to minimise aortocaval compression and maximise cardiac output
- Caesarean section should be performed after 4 minutes of unsuccessful Resuscitation
- Senior obstetric, anaesthetic and neonatal staff should be involved as early as possible
- Record keeping should be meticulous ensuring that treatment given and timings are clearly identified

3.2 Possible causes of cardiac arrest at term

a) Obstetric causes :

i. Massive haemorrhage

ii. Amniotic fluid embolism

iii. Eclampsia / HELLP syndrome

iv. Peripartum cardiomyopathy

b) Non-obstetric causes :

i. Pulmonary embolism

ii. Septic shock

iii. Heart failure

iv. Myocardial infarction

v. Trauma

vi. Anaesthetic complications

3.3 Resuscitation in Pregnancy

- In the event of maternal cardiorespiratory arrest, resuscitation should begin immediately and should follow current basic and advanced life support guidelines (Resuscitation Council UK). Physiological changes of pregnancy and the presence of the fetus demand some additions to the normal algorithms including :

    i. 30º left lateral tilt of the mother

    ii. Early tracheal intubation

    iii. Perimortem Caesarean section

3.4 Physiological Changes in Pregnancy

- The major physiological and anatomical changes that make resuscitation in pregnancy difficult include :

    i. Compression of the aorta and inferior vena cava by the uterus, decreases cardiac output by 25-30% in the supine position

    ii. In late pregnancy the uterus receives approximately 10% of the cardiac output.

- Therefore **tilt the mother approximately 30° to the left in order to reduce aortocaval compression**. This can be achieved by placing pillows or a Cardiff wedge under the patient's right side, or alternatively moving the uterus to the left by manual displacement or by raising the right hip.

- Cardiac compressions produce a maximum of 30% of the normal cardiac output[3] in non-pregnant patients in the supine position. When a patient is tilted laterally the efficiency of compressions is reduced further. However, the detrimental effect of aortocaval compression is greater than that of left tilt in maternal cardiac arrest.

- Maternal cardiac output increases by 40-50% in late pregnancy to satisfy the oxygen demands of the fetoplacental unit. Also, oxygen storage is reduced due to a decreased functional residual capacity of the lungs. As a result hypoxia occurs very

rapidly, and therefore early endotracheal intubation is helpful. However, attempts at intubation should not override oxygen delivery.

- Changes in gastric emptying and sphincter tone increase the risk of regurgitation and aspiration of stomach contents into the lungs. Therefore intubation will also protect the airway from aspiration of gastric contents.

3.5 Perimortem Caesarean Section

The concept of perimortem Caesarean section was introduced in 1986. The thinking behind it is that resuscitation is ineffective in the third trimester because of aortocaval compression, and therefore timely delivery will optimise outcome for mother and baby[4]. **Perimortem Caesarean section should therefore be initiated within 4 minutes of cardiac arrest if resuscitation is unsuccessful**, in order that cardiac output may be re-established within 5 minutes. This will minimise the danger of hypoxic neurological damage to the mother. Surgical packs should be available in the emergency department, on the delivery suite and on the antenatal ward to undertake a perimortem Caesarean Section, so that transfer to theatre is not necessary until after the baby is delivered. The neonatal resuscitation team should be present at delivery.

3.6 Further Care

The neonatal team will take charge of the baby. Early involvement of critical care specialists is essential, and the mother should be transferred to ICU for level 3 critical care. Relatives should be kept fully informed of events by senior staff. Records should be reviewed to make sure they are complete and any further retrospective information should be added once the patient is stable.

3.7 Unsuccessful Resuscitation

In the event of unsuccessful resuscitation the bereavement team should be involved. The head of midwifery and the clinical director should also be informed.

3.8 Staff Debrief

All staff involved in a maternal cardiac arrest should be involved in a formal debrief. This should be organised by senior midwifery and medical staff.

4. References

1. Donald F. Cardiopulmonary Resuscitation In Pregnancy Guideline (2006) Directorate of Women and Children's Health North Bristol N.H.S Trust

2. Chesnut DH. Obstetric Anaesthesia: Principles and Practice

3. American Heart Association. ACLS Provider Manual and BLS Manual, 2001

4. Sanders AB, Meislin HW, Ewy GA. The physiology of cardiopulmonary resuscitation. JAMA 1984; 252:328

5. Katz V, Balderston K, DeFreest M. Perimortem caesarean delivery: Were our assumptions correct? Am J Obs & Gyne (2005) 192,1916-21